Process Specification — Machining Inconel 718

Rev E · effective 2025-04-22. Applies to all parts using `MAT-INC718` on the DMG Mori DMU 50 platform (`DMU-14` , `DMU-22`).


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--- METADATA ---
document_id: PLM-PS-INC718-0042
title: Process Specification — Machining Inconel 718
domain: process_specs
source_system: PLM
revision: Rev E
effective_date: 2025-04-22
owner: M. Tanaka, Process Engineering — Wichita
part_numbers: [45621-B]
operation_numbers: [OP20, OP30]
machine_ids: [DMU-14, DMU-22]
controller_models: [Fanuc 0i-TF]
fault_codes: []
materials: [MAT-INC718]
standards: [AS9100D]
supplier_ids: [SUP-AERO-METALS]
jurisdictions: []
classification: internal
---
```


1. Material context

Inconel 718 (UNS N07718, AMS 5662 bar) is a nickel-based superalloy with high strength at elevated temperature. It is highly work-hardening and abrasive on tooling. Use this process specification for any Inconel 718 work on the DMU-14 / DMU-22 platform; for the specific application to Part `45621-B` , see work instruction `PLM-WI-45621B-OP30-RevC` .

2. Allowed tooling

- Roughing: 4-flute solid carbide end mills, AlTiN coating, Ø 6–12 mm.
- Finishing: 6-flute ball-nose solid carbide, AlTiN coating.
- HSS or uncoated carbide is **not permitted**.

3. Cutting parameter envelope

Operation	Surface speed (m/min)	Feed per tooth (mm)	Max axial DOC	Max radial DOC
Roughing	30–45	0.06–0.10	2.0 mm	40% of tool Ø

Semi-finishing	40–55	0.05–0.08	0.6 mm	0.5 mm
Finishing	50–70	0.04–0.06	0.2 mm	0.3 mm

These ranges must be combined with the OEM machine limits in `PDM-MAN-DMU14-007` §2. Where the two conflict, the lower value wins.

4. Tool-wear limits

Indicator	Limit	Action
Flank wear (VB)	**VBmax = 0.20 mm**	**Change tool immediately. Do not extend life — Inconel 718 work-hardens, so a worn tool generates a hardened surface layer that propagates rapid wear on the replacement tool.**
Cumulative cut time per finishing tool	60 min	Mandatory tool change regardless of visible wear.
Audible cue	Tonal change ("ringing")	Pause feed; inspect; replace if VB approaching 0.20 mm.

> Tool wear on Inconel 718 manifests as program-side issues (surface-finish degradation, dimensional drift, audible cue) and as a flag in the `DMU14_TOOL_LIFE` log. Tool wear does **not** generate a Fanuc controller alarm; alarm 300, in particular, is a position-feedback fault (`PDM-MAN-FANUC-0iTF-001` §7.3), unrelated to tool condition.

5. Coolant

Through-spindle coolant at 20 bar minimum. Coolant is EcoCut 7700 (`ERP-SDS-ECOCUT7700`) at 8–10% concentration (refractometer check daily). Below 18 bar through-spindle pressure, do not run — risk of chip evacuation failure and rapid tool damage.

6. Supplier and incoming material

Inconel 718 bar is procured exclusively from `SUP-AERO-METALS` per incoming spec `ERP-MAT-INC718-SPEC` . Per-lot material certifications are required (AS9100D §8.5.6 traceability). Any production batch made from a new lot must have its certification on file before the lot enters the spindle.

Document control / sign-off

Role	Name	Signature	Date
Document Owner	M. Tanaka, Process Engineering — Wichita	(on file)	2025-04-22

Approver	M. Tanaka, Process Engineering — Wichita	(on file)	2025-04-22